ANALYSTLAB AFRICA

TRACK B — DATA SCIENCE

Airbnb Pricing, Location and Listing Characteristics Investigation

Week 1 Analytical Report | Databricks | pandas, Spark, SciPy

1. Executive summary

This report investigates Airbnb listing characteristics across ten cities using the supplied Listings.csv and Reviews.csv files. The primary research question is how advertised prices vary across room types within each city. Supporting analyses examine accommodation capacity and advertised price, and historical review activity and aggregate listing ratings. The analysis is deliberately conducted within each city because the Listings data does not contain a dedicated currency field.

The supplied Listings file contains 279,712 records and 33 columns. It has 279,712 unique listing IDs and no duplicate listing_id values. The Reviews file contains 5,373,143 records and four fields. A streamed review-file inspection found 160 repeated review_id values (160 extra occurrences beyond the first), 193,556 distinct listing IDs with reviews, and no review listing IDs without a matching listing record. There are 86,156 listings without a review record.

2. Research problem and objectives

The business problem is to understand how advertised listing prices and listing characteristics differ across room types and cities, and whether available listing attributes and review indicators show useful statistical relationships. The intended users are stakeholders evaluating Airbnb market patterns. The analysis aims to provide descriptive evidence, test selected associations and identify limitations that must be considered before making decisions.

Research questions

- Primary: How does advertised listing price vary across room types within each city?
- Supporting 1: Is accommodation capacity associated with advertised listing price within each city?
- Supporting 2: Is listing-level review activity associated with the available aggregate listing rating?
- Supporting 3: How do listing distribution and historical review activity vary across cities?

3. Data sources and inspection

Listings.csv: 279,712 records, 33 columns. Fields include listing_id, host_id, city, neighbourhood, district, property_type, room_type, accommodates, bedrooms, price, host indicators and listing-level review scores.

Reviews.csv: 5,373,143 records, four columns: listing_id, review_id, date and reviewer_id. The Reviews data dictionary screenshot defines these fields as Listing ID, Review ID, Review date and Reviewer ID.

The Listings data dictionary contains definitions for all 33 Listings fields. The Reviews file does not contain review text or individual review scores.

4. Data quality findings

Check	Observed result	Treatment / implication
Listings row count	279,712	Use as source population for listing-level summaries.
Listings columns	33	Definitions checked against supplied Listings data dictionary.
listing_id uniqueness	0 duplicate rows; 279,712 unique IDs	Suitable as listing primary key in supplied extract.
Missing district	242,700 (86.8%)	Avoid district-level conclusions without a completeness caveat.
Missing host_response_time / response_rate	128,782 (46.0%)	Report response indicators only for available records.
Missing host_acceptance_rate	113,087 (40.4%)	Report denominator and missingness.
Missing bedrooms	29,435 (10.5%)	Use a documented missing-data approach for analyses involving bedrooms.
Missing overall review rating	91,405 (32.7%)	Rating analysis uses only listings with a rating.
Non-positive price	113 zero-price records; no negative values observed	Exclude non-positive prices from price tests; retain and document raw data.
Reviews row count	5,373,143	Process in Spark/chunks, not as one pandas dataframe.
Repeated review_id	160 repeated ID values; 160 extra occurrences	Investigate; do not assume review_id alone is unique.
Review/listing relationship	193,556 listings with reviews; 0 unmatched review listing IDs	Many-to-one relationship from review records to listings.
Listings without review records	86,156	Treat absence of review records as no rows in this extract, not as a zero rating.

5. Exploratory analysis findings

Listing counts differ substantially by city. Paris has the largest count (64,690), followed by New York (37,012) and Sydney (33,630). Hong Kong has the smallest count (7,087). These are counts of supplied listing records, not market-wide population estimates.

There are 113 zero-price listings. After excluding non-positive prices for price-specific summaries, 279,599 listings remain. Price distributions are strongly right-skewed: across the pooled records the median is 150 and the 99th percentile is 6,500, but these pooled figures mix local currencies and should not be interpreted as a cross-city price benchmark.

Within-city medians differ by room type. For example, in Paris the median advertised prices are 82 for Entire place, 53 for Private room, 34 for Shared room and 162.5 for Hotel room. In New York the respective medians are 140, 60, 45 and 163. In Cape Town, the medians are 1,287.5 for Entire place, 622 for Private room, 282 for Shared room and 1,350.5 for Hotel room. These amounts are in each city's local currency; they must not be compared across cities as if they shared one currency.

Across all listings, the median review count is 3, while 86,156 listings have no review records. Among listings with at least one review record, the median count is 8 and the maximum is 891. Review counts represent historical review records and are not occupancy or booking counts.

5.1 Airbnb Listing Distribution Across Cities

The first visualisation examines the distribution of Airbnb listings across ten international cities.

The results indicate that Paris has the highest number of listings, with approximately 64,690 records, followed by New York (37,012) and Sydney (33,630). Rome and Rio de Janeiro also demonstrate relatively high listing representation, with 27,647 and 26,615 listings respectively.

Hong Kong has the lowest representation, with approximately 7,087 listings.

The findings reveal an uneven distribution of listings across the selected cities. Paris alone contributes a substantially larger number of observations than Hong Kong.

Interpretation: The variation in listing counts suggests differences in the representation of accommodation markets across the dataset. However, listing volume should not be interpreted as a direct measure of tourism demand, occupancy or profitability.

Implication for further analysis: The unequal sample sizes should be considered when comparing accommodation prices, room types and review activity across cities. Subsequent comparisons should examine city-level distributions rather than relying solely on pooled results.

5.2 Room Type Composition Across Airbnb Cities

The second visualisation examines the proportional distribution of Airbnb room types across ten international cities using a 100% stacked bar chart.

The findings indicate that entire-place accommodation dominates the listing composition in most cities. Paris has the highest proportion, at approximately 86%, followed by Cape Town (74%) and Rio de Janeiro (73%).

Hong Kong presents a contrasting pattern, with private rooms accounting for approximately 55% of listings, compared with approximately 37% for entire places. Private rooms also represent a substantial share of accommodation supply in Istanbul, Mexico City and New York.

Hotel rooms and shared rooms generally account for relatively small proportions of the available listings.

These findings demonstrate that accommodation supply differs considerably across the selected cities. The dominance of entire-place listings in some markets, compared with the stronger representation of private rooms in others, suggests differences in the composition of Airbnb accommodation offerings.

Business implication: Airbnb hosts and accommodation investors may benefit from understanding the dominant room-type composition within each market before making accommodation investment or positioning decisions.

Analytical consideration: These percentages describe the composition of the dataset rather than the entire accommodation market in each city. They should not be interpreted as direct measures of customer demand or booking preferences.

5.3 Advertised Price Distribution by Room Type Within Each City

The third visualisation investigates differences in advertised Airbnb prices across four accommodation categories: entire place, hotel room, private room and shared room.

Boxplots were generated separately for each city to facilitate comparisons between room types within individual markets. A logarithmic scale was applied to the price axis to improve the visualisation of positively skewed price distributions.

The results indicate that entire-place accommodation generally has higher median advertised prices than private and shared rooms. This pattern is particularly noticeable in Hong Kong, Istanbul, Rio de Janeiro, Rome and Sydney.

However, some markets demonstrate different pricing patterns. In Paris, Mexico City and New York, hotel rooms display higher median advertised prices than entire-place accommodation. Bangkok also shows relatively comparable median prices for hotel rooms and entire places.

Shared rooms generally have the lowest median advertised prices across the cities, consistent with their positioning as a lower-cost accommodation option.

The visualisations also reveal differences in price variability across accommodation categories, indicating that advertised prices are not uniform even within the same room type and city.

Business implication: Accommodation providers should consider local market conditions and room-type positioning when establishing pricing strategies. The findings also highlight the importance of evaluating accommodation categories separately rather than applying a single pricing assumption across all markets.

Limitation: The analysis uses advertised prices rather than actual transaction prices. Additionally, local currencies differ between cities, making direct numerical comparisons across cities inappropriate. The observed patterns are descriptive and require statistical testing to determine whether the differences are statistically significant.

5.4 Accommodation Capacity and Advertised Price

This visualisation examines the relationship between maximum guest accommodation capacity and advertised Airbnb listing prices across ten international cities.

The scatterplots suggest a generally positive relationship between accommodation capacity and advertised price. Larger properties tend to command higher advertised prices, although considerable variation exists among listings accommodating similar numbers of guests.

Sydney and Cape Town display particularly visible upward patterns, while Hong Kong and New York demonstrate substantial price dispersion across different accommodation capacities.

The analysis also reveals that listings accommodating smaller groups are considerably more common than those designed for larger groups. Several unusually high advertised prices are visible, highlighting the importance of investigating extreme observations during data preparation.

Business implication: Accommodation capacity appears to be an important consideration when evaluating Airbnb pricing strategies. However, property capacity alone does not explain advertised prices. Other factors, including location, property characteristics, amenities and room type, may also contribute to pricing differences.

The visual evidence is consistent with the positive Spearman correlations identified in the preliminary statistical analysis. Further statistical validation is required before drawing conclusions about the strength of these relationships.

5.5 Historical Airbnb Review Activity Over Time

This visualisation investigates historical Airbnb review activity using monthly aggregations of review records.

The chart reveals a substantial increase in review activity over the observed period. Monthly review volumes remain relatively low during the earlier years, followed by a stronger upward trend from approximately 2014 onwards.

Review activity increases considerably between 2016 and 2019, reaching a peak of approximately 168,000 monthly review records in late 2019.

A pronounced decline is visible during 2020, followed by a partial recovery. This period coincides with the global COVID-19 pandemic and associated travel disruptions. Although this provides relevant historical context, the visualisation alone cannot establish a causal relationship between the pandemic and declining review activity.

The final observations also show a substantial reduction in recorded reviews, potentially reflecting incomplete periods or dataset coverage limitations.

Business implication: Historical review activity provides useful insight into changes in customer engagement and the development of Airbnb's listing ecosystem. However, review volumes should not be interpreted as direct booking volumes or occupancy rates.

Further investigation into the dataset's reporting period and completeness is recommended before drawing conclusions about the final observations.

5.6 Historical Review Activity and Aggregate Listing Rating

This visualisation examines the relationship between historical review activity and overall Airbnb listing ratings.

The scatterplot shows that most listings with available rating information achieve relatively high overall ratings, with a substantial concentration of observations between 90 and 100.

However, the distribution does not reveal an obvious strong positive relationship between the number of historical reviews and aggregate listing ratings. Listings with relatively few reviews can achieve high ratings, while listings with larger review histories also display variation in their ratings.

The chart also highlights greater rating variability among listings with lower review counts. This observation warrants further investigation through statistical analysis.

The concentration of observations near the upper rating boundary suggests a ceiling effect, which may limit the ability to distinguish between highly rated listings.

Business implication: Review volume alone should not be treated as an indicator of customer satisfaction. Airbnb hosts and platform managers should consider both review engagement and the actual ratings received when evaluating listing performance.

A Spearman rank correlation test is recommended to quantify the strength and direction of the association between historical review counts and aggregate listing ratings.

6. Hypotheses and statistical methods

H1 — Price distributions differ by room type within a city

H0: Within a city, advertised price distributions are the same across room types. H1: At least one room type has a different price distribution. A Kruskal–Wallis H test was run separately for each city, including room-type groups with at least 30 valid positive-price observations. This avoids pooling prices across currencies and is suitable for skewed, non-normal price data.

City	N tested	H statistic	Epsilon-squared
Bangkok	19,360	932.98	0.048
Cape Town	19,086	2,689.31	0.141
Hong Kong	7,087	1,889.27	0.266
Istanbul	24,517	5,308.34	0.216
Mexico City	20,058	6,586.43	0.328
New York	36,984	15,303.15	0.414
Paris	64,628	5,339.01	0.083
Rio de Janeiro	26,608	4,062.89	0.153
Rome	27,641	4,243.48	0.153
Sydney	33,630	14,359.65	0.427

The Kruskal–Wallis test is statistically significant in all ten cities ($p < 0.001$ in each case). Effect-size estimates (epsilon-squared) range from approximately 0.048 in Bangkok to 0.427 in Sydney. This supports differences in at least one room-type distribution within each city. The omnibus test does not identify which specific pairs differ; pairwise post-hoc testing with multiplicity correction would be needed for that.

H2 — Accommodation capacity is associated with advertised price

H0: Spearman's rank correlation between accommodates and price is zero within a city. H1: The correlation is non-zero. Spearman correlation was calculated separately by city.

City	N	Spearman rho	p-value
Bangkok	19,360	0.313	<0.001

Cape Town	19,086	0.671	<0.001
Hong Kong	7,087	0.513	<0.001
Istanbul	24,517	0.506	<0.001
Mexico City	20,058	0.557	<0.001
New York	36,984	0.554	<0.001
Paris	64,628	0.529	<0.001
Rio de Janeiro	26,608	0.518	<0.001
Rome	27,641	0.494	<0.001
Sydney	33,630	0.680	<0.001

All ten cities show positive, statistically significant rank associations ($p < 0.001$). Correlations range from 0.313 in Bangkok to 0.680 in Sydney. This indicates that listings accommodating more guests tend to have higher advertised prices within each city. The relationship is observational and does not prove that capacity causes higher prices.

H3 — Review activity is associated with aggregate listing rating

Among 188,307 listings with an available overall rating, the Spearman correlation between review count and overall listing rating is rho = -0.205 ($p < 0.001$). The association is negative and small in magnitude. This does not establish that receiving more reviews reduces ratings. Listing age, exposure, market mix and other unobserved factors may influence both measures. Review volume is not a measure of occupancy or guest sentiment.

7. Required visualisations

- Unique listing counts by city.
- Advertised price distributions by room type, faceted by city and shown on a log scale.
- Room-type composition by city.
- Accommodation capacity versus advertised price, by city.
- Monthly historical review activity.
- Historical review count versus aggregate listing rating.

The Databricks notebook includes executable code for all six visualisations. The final notebook run should be saved with the rendered plots and result tables as evidence.

8. Limitations

- Currency: the Listings data has no dedicated currency field. Raw prices are analysed within each city only.
- Missingness: district, host response indicators, bedrooms and review scores contain missing values.
- Repeated review identifiers: 160 review_id values repeat. The project specification says the composite listing_id + review_id is unique; verify this in Databricks before any deduplication.
- Review content: there are no written review comments or individual review ratings, so sentiment analysis is unsupported.
- Unequal sample sizes: city listing counts differ, affecting precision and representativeness.
- Extreme prices: high values can distort means; medians and interquartile ranges are preferred.
- No bookings or occupancy: the supplied files do not establish actual bookings, occupancy, revenue or profit.
- Association is not causation: the analyses are observational.

9. Recommendations

1. Use within-city room-type median prices and interquartile ranges to compare listing segments. Do not rank cities using raw price amounts.
2. Use accommodation capacity as a useful segmentation variable, while considering property type, bedrooms and location where data completeness allows.
3. Display review volume and aggregate ratings as separate indicators. Do not label review count as occupancy or rating as review-text sentiment.
4. Improve completeness of district, bedroom and host response/acceptance fields; report denominators and missingness with every KPI.
5. Preserve raw data and document all cleaning rules, especially the treatment of zero prices and repeated review identifiers.
6. For future analysis, add explicit currency metadata, booking/occupancy records and review text or item-level review scores only if properly authorised and available.

10. Success criteria

- All 279,712 listing records are ingested and listing_id uniqueness is validated.
- The Reviews dataset is processed without loading all 5.37 million rows into pandas.
- The listing-to-review relationship is validated and repeated review IDs are documented.
- At least five visualisations are produced from reproducible code.
- At least three hypotheses are tested with suitable methods, sample sizes, p-values and interpretation.
- All price comparisons are within city unless currency conversion is separately documented.
- The report distinguishes review activity, aggregate ratings, advertised prices and actual booking/revenue outcomes.